

**Vidya Vikas Institute of Engineering and Technology,
Mysore
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VTU Question Papers

“Computer Science Engineering”

IV, VI & VIII Semester

June/July 2023

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

INDEX

S/L	Subject Code	Subject Title	Exam Date
1	21BE45	Biology for Engineers	June/July 2023
2	21CIP37/21CIP47	CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS	June/July 2023
3	21CS42	Design & Analysis of Algorithms	June/July 2023
4	21CS43	Microcontroller and Embedded Systems	June/July 2023
5	21CS44	Operating System	June/July 2023
6	21KBK47	Balake Kannada	June/July 2023
7	21KSK47	Samskruthika Kannada	June/July 2023
8	21UH49	Universal Human Values	June/July 2023
9	18CS61	System Software and Compilers	June/July 2023
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11	1 8CS63	Web Technology and its Application	June/July 2023
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13	18CS651	Mobile Application Development	June/July 2023
14	18CS653	Programming in Java	June/July 2023
15	18CS654	Introduction to Operating System	June/July 2023
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17	18CS822	Storage Area Networks	June/July 2023

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023

Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is a biomolecule? Explain the classification of biomolecule.	07	L2	CO1
	b.	Explain the role of DNA vaccine for rabies and RNA vaccine for COVID-19.	07	L2	CO1
	c.	Write a short note on cellulose based bio-filters.	06	L2	CO1
OR					
Q.2	a.	Explain the DNA finger printing in forensic applications.	07	L2	CO1
	b.	Explain the role of lipids and its application in cleaning agents.	07	L2	CO1
	c.	Write a short note on biosensors and bioplastics.	06	L2	CO1
Module – 2					
Q.3	a.	Explain brain as a CPU system.	07	L3	CO1
	b.	Explain eye as a camera system.	07	L3	CO1
	c.	Write a short note on cardiac pacemaker.	06	L2	CO1
OR					
Q.4	a.	Explain the robotic arms for prosthetics.	07	L3	CO1
	b.	Explain heart as a pump system.	07	L3	CO1
	c.	Write a short note on engineering solutions for Parkinson's disease.	06	L2	CO1
Module – 3					
Q.5	a.	Explain the lungs as a purification system.	07	L3	CO2
	b.	Explain the kidney as filtration system.	07	L3	CO2
	c.	Write a short note on spirometry and ventilator.	06	L2	CO2
OR					
Q.6	a.	Explain muscular and skeletal system as scaffolds.	07	L3	CO2
	b.	Explain bio-engineering solutions for muscular dystrophy and osteoporosis.	07	L3	CO2
	c.	Write a short note on Chronic Obstructive Pulmonary Disease (COPD).	06	L2	CO2
Module – 4					
Q.7	a.	Explain the terms Echolocation Ultrasonography and Sonars.	07	L3	CO3
	b.	Explain the process of Photosynthesis and Photovoltaic cells.	07	L3	CO3
	c.	Write a short note on Bionic leaf, GPS, Bird flight and aircraft.	06	L2	CO3
OR					
Q.8	a.	Explain the terms Lotus leaf effect, Plant Burrs and Super hydrophobic and self-cleaning surfaces.	07	L3	CO3
	b.	Explain the terms Spark skin and Swimsuits, Bullet train using biological concepts.	07	L3	CO3
	c.	Write a short note on Hemoglobin – Based Oxygen Carriers (HBOC's) and Perfluorocarbons (PFC).	06	L2	CO3
Module – 5					
Q.9	a.	Explain the DNA Organic and Biocomputing.	07	L3	CO4
	b.	Explain the Bioimaging and Artificial intelligence for Disease Diagnosis.	07	L3	CO4
	c.	Write a short note on Self healing Bioconcrete.	06	L2	CO4
OR					
Q.10	a.	Explain the importance of Bioimaging.	07	L3	CO4
	b.	Explain Bioremediation and Bio-Mining via microbial surface adsorption.	07	L3	CO4
	c.	Write a short note on Nanomedicines and Bioleaching.	06	L2	CO4

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Question Paper Version : A

Third/Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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1. When was the Indian constitution enacted and adopted?
a) 26/10/1949 b) 26/11/1949 c) 26/4/1949 d) 26/01/1950
 2. 'We the people of India' are the opening words of the,
a) Preamble of the Indian constitution b) Article 21 of the Indian constitution
c) Fundamental rights d) Directive principles of state policy
 3. Which one of these is the primary source of the Indian constitution?
a) British constitution b) Irish constitution
c) Charter Act of 1833 d) Government of India Act of 1935
 4. The original Indian constitution had :
a) 12 parts, 6 schedule and 320 Articles b) 20 parts, 8 schedule and 380 Articles
c) 12 parts, 8 schedule and 396 Articles d) 12 parts, 10 schedule and 300 Articles
 5. The word 'Sovereign' means that,
a) Supreme in nature b) A country is under dictatorship
c) A country is poor of weak d) A country is strong and powerfull
 6. Directive principles are,
a) Justiciable b) Not practiced at rural levels
c) Non-justiciable d) Associated to the Government worker's
 7. How much time was taken for training the constitution?
a) 1 year, 11 months, 18 days b) 5 year, 11 months, 18 days
c) 2 year, 11 months, 18 days d) 3 year, 11 months, 18 days

8. India is a Sovereign, socialist, secular, democratic and republic in the Indian constitution this expression occurs in,
a) Citizenship
b) Preamble
c) Fundamental rights
d) Directive principles
9. Who among the following is the supreme commander of the Armed forces?
a) Air Chief Marshal
b) Prime Minister
c) Defense Minister
d) President
10. The 91st Amendment Act (2003) is associated with,
a) Size of the council of ministers
b) Primary education
c) Fundamental Duty
d) Powers of the President
11. How many members were nominated to the parliament by the president of India?
a) 14 members
b) 12 members
c) 2 members
d) 6 members
12. Who among the following distribute portfolios for the council of minister,
a) President
b) Vice president
c) Prime Minister
d) Speaker of Lok Sabha
13. The chief justice and other judges of the supreme court hold office till they complete,
a) Sixty years
b) Sixty five years
c) Sixty two years
d) Seventy years
14. The council of ministers are responsible to the,
a) Rajya Sabha
b) Vidhan Parishad
c) Lok Sabha
d) Supreme court
15. The Vice-President of India is elected by the,
a) Judges of the supreme court
b) President
c) Prime Minister
d) Members of parliament
16. Who can issue ordinance when the parliament is not in session:
a) President
b) High court judges
c) Home minister
d) Finance minister
17. In case of the violation of the Fundamental Rights we may approach the,
a) Civil Courts
b) Supreme Court
c) High Court
d) Both (a) and (b)
18. Which of the following equalities is/are included in the Right to Equality?
a) Equality before law
b) Equal protection of law
c) Equal opportunities in the public employment
d) All of these.
19. Prohibition of trafficking in human beings and forced labour comes under which of the following fundamental right?
a) Right to freedom
b) Right against exploitation
c) Cultural & Educational Right
d) Right to equality.
20. There is no provision in the constitution for the impeachment of the,
a) President
b) Vice President
c) Governor
d) Supreme court Judges
21. The tenure of the Council of state is,
a) Not subject to dissolution
b) 2 years
c) 5 years
d) 4 years
22. When elections are held in one or a few constituencies due to death or resignation of candidates, it is called as _____.
a) General election
b) Primary election
c) By election
d) Midterm election

23. Fundamental Rights are borrowed from the constitution of,
a) UK b) USA c) Germany d) Ireland
24. What is the minimum age to become Judges of Supreme Court of India?
a) 25 years b) 30 years c) 35 years d) None of these
25. The Indian constitution gives the power of amending the constitution to,
a) The people of India b) The president
c) The Parliament d) Supreme Court of India
26. Right to Education (RTE) was introduced in _____Amendment,
a) 86th b) 42nd c) 44th d) 61st
27. How many types of writs can be issued by the Supreme Court for the protection of Fundamental Rights?
a) Four b) Five c) One d) Six
28. Who presides over the sessions of Rajya Sabha?
a) Speaker b) Home minister c) Vice-president d) President
29. Who appoints the Vice-Chancellors of the state universities?
a) Education minister b) District commissioner c) Chief minister d) Governor
30. Election commission is a _____ body and the term of election commission is _____years or _____years of age whichever is earlier.
a) Uni-member, 4 years or 62 years b) Multi-member, 6 years or 65 years
c) Constitutional body, 5 years or 60 years d) None of these
31. The member to be nominated by the President for the council of states are from,
a) Literature b) Science c) Sports d) All of these
32. Which of the following Pairs is not property matched ,
a) 44th Amendment-citizenship act b) 52nd Amendment-Anti Defection Law
c) 42nd Amendment-Fundamental duties d) 73rd Amendment-Local self Government
33. The speaker of Lok Sabha,
a) is appointed by the President b) is nominated by the Vice-President
c) is chosen by the members of Lok Sabha d) is elected by the members of parliament
34. Financial Emergency has been imposed in India,
a) Once b) Never c) Twice d) Thrice
35. Respect for the National Flag and the National Anthem is,
a) a Fundamental right b) a Fundamental Duty
c) a Directive principle d) an ordinary duty
36. A non-member of the state legislature can be the minister for a period not exceeding,
a) Six month b) One year c) Six weeks d) Three months
37. Engineering ethics is a,
a) developing ethics b) Preventive ethics
c) natural ethics d) Scientifically developed ethics

38. Risk estimation can be done by using,
a) Cooking b) Trimming c) Event tree d) Both (a) and (b)
39. The Patent holder does not allow others to use patented information for _____ years from the date of filing.
a) 25 b) 30 c) 50 d) 20
40. The use of intellectual property of others without their permission or credit is referred to as,
a) Cooking b) Plagiarism c) Patents d) Formulae
41. Which of the following is not the concept of responsibilities?
a) Minimalist b) Reasonable care c) Utilitarianism d) Good works
42. Lying means,
a) Intentionally conveying false information to others b) Fabrication
c) Plagarism d) All of these
43. The three types of Justice reffered in our preamble are :
a) Social, Economic and Religious b) Social, Economic and Natural
c) Social, Economic and International d) Social, Economic and Political
44. An arrested person must be produced before a magistrate within _____hours of arrest.
a) 12 b) 24 c) 36 d) 48
45. Election commission conducts the election as per which act?
a) Parliament act b) People's representative act of 1982
c) Code of conduct act d) State representative act
46. When the office of the president, falls vacant, the same must be filled up with in?
a) 3 months b) 6 months c) 1 year d) 9 months
47. Who among the following are not entitled to form Union or Association,
a) Police b) Teachers c) Workers d) Doctors
48. The MLA's of various state legislative assemblies are varying between,
a) 40 to 450 b) 50 - 500 c) 28 - 12 d) 60 - 500
49. A bill cannot become an act of parliament, unless and until _____?
a) it is passed by Lok Sabha b) it is passed by Rajya Sabha
c) it gets assent from President d) it gets approved by Supreme Court
50. Who hoisted the National Flag during 74th Republic day function in New Delhi?
a) Prime Minister b) President
c) Vice-President d) Chief justice of India

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Question Paper Version : B**Third/Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023****CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS**

[Time: 1 hrs.]

[Max. Marks: 50]

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Question Paper Version : C

Third/Fourth Semester B.E. /B.Tech. Degree Examination, June/July 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

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a) Sixty years b) Sixty five years c) Sixty two years d) Seventy years
44. The council of ministers are responsible to the,
a) Rajya Sabha b) Vidhan Parishad c) Lok Sabha d) Supreme court
45. The Vice-President of India is elected by the,
a) Judges of the supreme court b) President
c) Prime Minister d) Members of parliament
46. Who can issue ordinance when the parliament is not in session:
a) President b) High court judges c) Home minister d) Finance minister
47. In case of the violation of the Fundamental Rights we may approach the,
a) Civil Courts b) Supreme Court c) High Court d) Both (a) and (b)
48. Which of the following equalities is/are included in the Right to Equality?
a) Equality before law b) Equal protection of law
c) Equal opportunities in the public employment d) All of these.
49. Prohibition of trafficking in human beings and forced labour comes under which of the following fundamental right?
a) Right to freedom b) Right against exploitation
c) Cultural & Educational Right d) Right to equality.
50. There is no provision in the constitution for the impeachment of the,
a) President b) Vice President c) Governor d) Supreme court Judges

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Question Paper Version : D

Third/Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The tenure of the Council of state is,
a) Not subject to dissolution b) 2 years c) 5 years d) 4 years
 2. When elections are held in one or a few constituencies due to death or resignation of candidates, it is called as _____.
a) General election b) Primary election
c) By election d) Midterm election
 3. Fundamental Rights are borrowed from the constitution of,
a) UK b) USA c) Germany d) Ireland
 4. What is the minimum age to become Judges of Supreme Court of India?
a) 25 years b) 30 years c) 35 years d) None of these
 5. The Indian constitution gives the power of amending the constitution to,
a) The people of India b) The president
c) The Parliament d) Supreme Court of India
 6. Right to Education (RTE) was introduced in _____ Amendment,
a) 86th b) 42nd c) 44th d) 61st
 7. How many types of writs can be issued by the Supreme Court for the protection of Fundamental Rights?
a) Four b) Five c) One d) Six
 8. Who presides over the sessions of Rajya Sabha?
a) Speaker b) Home minister c) Vice-president d) President

9. Who appoints the Vice-Chancellors of the state universities?
a) Education minister b) District commissioner c) Chief minister d) Governor
10. Election commission is a _____ body and the term of election commission is _____ years or _____ years of age whichever is earlier.
a) Uni-member, 4 years or 62 years b) Multi-member, 6 years or 65 years
c) Constitutional body, 5 years or 60 years d) None of these
11. Which of the following is not the concept of responsibilities?
a) Minimalist b) Reasonable care c) Utilitarianism d) Good works
12. Lying means,
a) Intentionally conveying false information to others b) Fabrication
c) Plagiarism d) All of these
13. The three types of Justice referred in our preamble are :
a) Social, Economic and Religious b) Social, Economic and Natural
c) Social, Economic and International d) Social, Economic and Political
14. An arrested person must be produced before a magistrate within _____ hours of arrest.
a) 12 b) 24 c) 36 d) 48
15. Election commission conducts the election as per which act?
a) Parliament act b) People's representative act of 1982
c) Code of conduct act d) State representative act
16. When the office of the president, falls vacant, the same must be filled up with in?
a) 3 months b) 6 months c) 1 year d) 9 months
17. Who among the following are not entitled to form Union or Association,
a) Police b) Teachers c) Workers d) Doctors
18. The MLA's of various state legislative assemblies are varying between,
a) 40 to 450 b) 50 - 500 c) 28 - 12 d) 60 - 500
19. A bill cannot become an act of parliament, unless and until _____?
a) it is passed by Lok Sabha b) it is passed by Rajya Sabha
c) it gets assent from President d) it gets approved by Supreme Court
20. Who hoisted the National Flag during 74th Republic day function in New Delhi?
a) Prime Minister b) President
c) Vice-President d) Chief justice of India
21. The member to be nominated by the President for the council of states are from,
a) Literature b) Science c) Sports d) All of these
22. Which of the following Pairs is not properly matched ,
a) 44th Amendment-citizenship act b) 52nd Amendment-Anti Defection Law
c) 42nd Amendment-Fundamental duties d) 73rd Amendment-Local self Government
23. The speaker of Lok Sabha,
a) is appointed by the President b) is nominated by the Vice-President
c) is chosen by the members of Lok Sabha d) is elected by the members of parliament

24. Financial Emergency has been imposed in India,
a) Once b) Never c) Twice d) Thrice
25. Respect for the National Flag and the National Anthem is,
a) a Fundamental right b) a Fundamental Duty
c) a Directive principle d) an ordinary duty
26. A non-member of the state legislature can be the minister for a period not exceeding,
a) Six month b) One year c) Six weeks d) Three months
27. Engineering ethics is a,
a) developing ethics b) Preventive ethics
c) natural ethics d) Scientifically developed ethics
28. Risk estimation can be done by using,
a) Cooking b) Trimming c) Event tree d) Both (a) and (b)
29. The Patent holder does not allow others to use patented information for _____ years from the date of filing.
a) 25 b) 30 c) 50 d) 20
30. The use of intellectual property of others without their permission or credit is referred to as,
a) Cooking b) Plagiarism c) Patents d) Formulae
31. How many members were nominated to the parliament by the president of India?
a) 14 members b) 12 members c) 2 members d) 6 members
32. Who among the following distribute portfolios for the council of minister,
a) President b) Vice president c) Prime Minister d) Speaker of Lok Sabha
33. The chief justice and other judges of the supreme court hold office till they complete,
a) Sixty years b) Sixty five years c) Sixty two years d) Seventy years
34. The council of ministers are responsible to the,
a) Rajya Sabha b) Vidhan Parishad c) Lok Sabha d) Supreme court
35. The Vice-President of India is elected by the,
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38. Which of the following equalities is/are included in the Right to Equality?
a) Equality before law b) Equal protection of law
c) Equal opportunities in the public employment d) All of these.
39. Prohibition of trafficking in human beings and forced labour comes under which of the following fundamental right?
a) Right to freedom b) Right against exploitation
c) Cultural & Educational Right d) Right to equality.

40. There is no provision in the constitution for the impeachment of the,
a) President b) Vice President c) Governor d) Supreme court Judges
41. When was the Indian constitution enacted and adopted?
a) 26/10/1949 b) 26/11/1949 c) 26/4/1949 d) 26/01/1950
42. 'We the people of India' are the opening words of the,
a) Preamble of the Indian constitution b) Article 21 of the Indian constitution
c) Fundamental rights d) Directive principles of state policy
43. Which one of these is the primary source of the Indian constitution?
d) British constitution b) Irish constitution
c) Charter Act of 1833 d) Government of India Act of 1935
44. The original Indian constitution had :
a) 12 parts, 6 schedule and 320 Articles b) 20 parts, 8 schedule and 380 Articles
c) 12 parts, 8 schedule and 396 Articles d) 12 parts, 10 schedule and 300 Articles
45. The word 'Sovereign' means that,
a) Supreme in nature b) A country is under dictatorship
c) A country is poor of weak d) A country is strong and powerfull
46. Directive principles are,
a) Justiciable b) Not practiced at rural levels
c) Non-justiciable d) Associated to the Government worker's
47. How much time was taken for training the constitution?
a) 1 year, 11 months, 18 days b) 5 year, 11 months, 18 days
c) 2 year, 11 months, 18 days d) 3 year, 11 months, 18 days
48. India is a Sovereign, socialist, selular, democratic and republic in the Indian constitution this expression occurs in,
a) Citizenship b) Preamble
c) Fundamental rights d) Directive principles
49. Who among the following is the supreme commander of the Armed forces?
a) Air Chief Marshal b) Prime Minister
c) Defense Minister d) President
50. The 91st Amendment Act (2003) is associated with,
a) Size of the council of ministers b) Primary education
c) Fundamental Duty d) Powers of the President

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Fourth Semester B.E. Degree Examination, June/July 2023

Design & Analysis of Algorithms

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the algorithm design and analysis process in detail. (10 Marks)
- b. Explain the asymptotic narrations with example. (10 Marks)

OR

- 2 a. Explain the general plan of mathematical analysis of recursive algorithm with example. (10 Marks)
- b. Design an algorithm to search an element in an array using sequential search. Discuss the Best-case, worst-case and average-case efficiency of this algorithm. (10 Marks)

Module-2

- 3 a. Explain the concept of Divide and Conquer. Write the recursive algorithm to perform Binary search on the list of elements. (10 Marks)
- b. Apply Quick sort algorithm to sort the list of characters : P, R, O, G, R, A, M, M, I, N, G. Draw the tree of recursive calls made while tracing. (10 Marks)

OR

- 4 a. Develop a recursive algorithm to find the minimum and maximum element from the list. Illustrate with an example. (10 Marks)
- b. Define Topological sorting. Illustrate the topological sorting for the following graph:

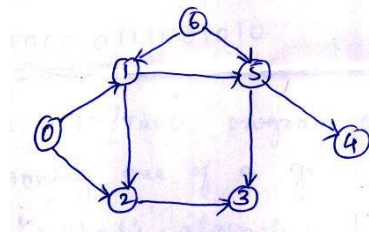


Fig. Q4 (b)

(10 Marks)

Module-3

- 5 a. Solve the following instance of greedy knapsack problem where $n = 4$, $m = 10$, $p = \{40, 42, 25, 12\}$ and $w = \{4, 7, 5, 3\}$. (10 Marks)
- b. Apply Dijkstra's algorithm to find single source shortest path for the given graph by considering 'S' as the source vertex. (10 Marks)

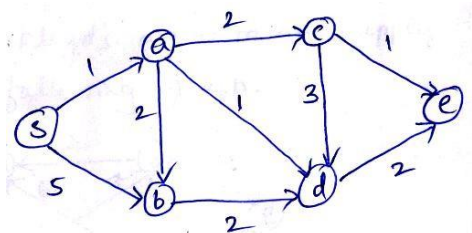


Fig. Q5 (b)

OR

- 6 a. Construct a Huffman Tree and resulting code word for the following :

Character	A	B	C	D	-
Probability	0.4	0.1	0.2	0.15	0.15

Encode the text ABACABAD and Decode the text 100010111001010.

(10 Marks)

- b. Write a C++/Java program to find minimum cost spanning tree of a given connected graph using Kruskal's algorithm. Use Union-Find algorithm in your program. (10 Marks)

Module-4

- 7 a. Find a minimum-cost path from S to T in the given multistage graph.

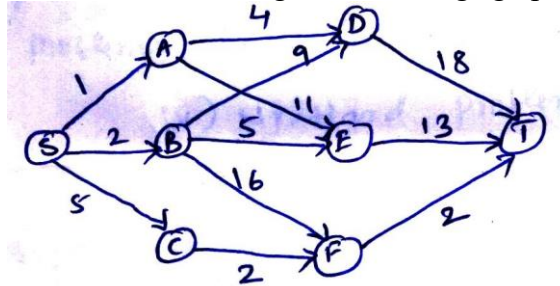


Fig. Q7 (a)

(10 Marks)

- b. Write Floyd's algorithm and apply the same to trace the following graph.

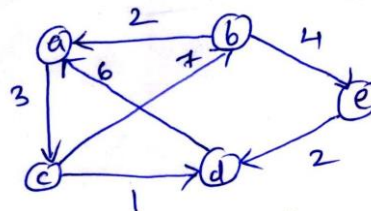


Fig. Q7 (b)

(10 Marks)

OR

- 8 a. Write Horspool's algorithm for string matching. Find the pattern BARBER. In the text : JIM_SAW_ME_IN_A_BARBERSHOP. (10 Marks)
- b. Write a C++/Java program to solve 0/1 knapsack problem using Dynamic programming method. (10 Marks)

Module-5

- 9 a. Differentiate between Back tracking and Branch and Bound technique. Apply back tracking to solve the following instance of the subset-sum problem : $S = \{1, 2, 3, 6, 8\}$, $d = 9$. (10 Marks)
- b. Solve the following assignment problem using branch and bound method. (10 Marks)

	Job 1	Job 2	Job 3	Job 4
Person a	9	2	7	8
Person b	6	4	3	7
Person c	5	8	1	8
Person d	7	6	9	4

OR

- 10 a. Explain the following with examples :
 (i) P problems (ii) NP problems
 (iii) NP-complete problems (iv) NP-Hard problems (10 Marks)
- b. Design and implement C++/Java program to find all Hamiltonian cycles in a connected undirected graph G of n vertices using back tracking principle. (10 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2023 Microcontroller and Embedded Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Mention the differences between:
(i) Microprocessor and Microcontrollers (ii) CISC and RISC (10 Marks)
b. With a neat diagram, explain embedded system hardware. (10 Marks)

OR

- 2 a. Explain in detail about Current Program Status Register (CPSR). (10 Marks)
b. With a neat diagram, explain embedded system software. (10 Marks)

Module-2

- 3 a. Explain different branch instruction in ARM processor. (10 Marks)
b. Discuss different types of addressing modes for load store multiple instructions with example. (10 Marks)

OR

- 4 a. Explain single register load store addressing mode syntax, table, index mode with an example. (10 Marks)
b. Discuss SWAP instruction with an example. (10 Marks)

Module-3

- 5 a. Write a C program that prints the square of the integers between 0 to 9 using functions and explain how to convert this C function to an assembly function with command. (10 Marks)
b. Discuss how instruction is scheduled in ARM. (10 Marks)

OR

- 6 a. Explain code optimization, profiling and cycle counting. (10 Marks)
b. Discuss how Registers are allocated to optimize the program. (10 Marks)

Module-4

- 7 a. Explain the purpose of embedded systems used in various domains. (10 Marks)
b. Write a note on core of the embedded systems. (10 Marks)

OR

- 8 a. Explain different classification of embedded system. Give an example for each. (10 Marks)
b. Write a note on sensors and actuators used in various embedded systems. (10 Marks)

Module-5

- 9 a. Explain multi threading. (06 Marks)
b. Explain the concept of deadlock with a neat diagram. (04 Marks)
c. Write a note on message passing. (10 Marks)

OR

- 10 a. Write a note on multiprocessing and multi-tasking. (10 Marks)
b. Explain the role of Integrated Development Environment (IDE) for Embedded Software Development. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, 42+8 = 50, will be treated as malpractice.

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Fourth Semester B.E. Degree Examination, June/July 2023
Operating System

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With neat diagram, explain :
 i) Operating System ii) Dual Mode Operation in O.S. **(08 Marks)**
 b. Explain the Operating System services with respect to programs and the users. **(06 Marks)**
 c. What is a Process? Explain different states of a process with state diagram. **(06 Marks)**

OR

- 2 a. With neat diagram, explain the concept of Non virtual machine, Virtual machine and VMware architecture. **(10 Marks)**
 b. Define : i) Context switching ii) Direct and Indirect communication
 iii) Automatic and Explicit buffering. **(10 Marks)**

Module-2

- 3 a. What is Multithreaded process? Explain four benefits of Multithreaded programming. **(06 Marks)**
 b. Consider the set of process with Arrival time, CPU burst time (in milliseconds) and priority shown below (Lower number represents higher priority).

Process	Arrival Time	Burst Time	Priority
P1	0	10	3
P2	1	1	1
P3	2	2	4
P4	3	1	5
P5	4	5	2

Write the Gantt chart and solve the Average waiting time and Average turnaround time for

i) SJF Scheduling (Preemptive) ii) Priority Scheduling (Preemptive).

(NOTE : Consider Arrival Time for both Algorithms).

(14 Marks)**OR**

- 4 a. Explain with diagram : i) Multithreading models ii) Multilevel queue scheduling. **(08 Marks)**
 b. What is Critical – Section? How do you implement a monitor solution to the dining – philosophers problem? **(12 Marks)**

Module-3

- 5 a. What is a Deadlock? What are the four necessary conditions for the deadlock to occur? **(04 Marks)**
 b. What are the two methods to eliminate deadlock? **(02 Marks)**

- c. Consider the following snapshot of a system :

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P1	2	0	0	1	4	2	1	2	3	3	2	1
P2	3	1	2	1	5	2	5	2				
P3	2	1	0	3	2	3	1	6				
P4	1	3	1	2	1	4	2	4				
P5	1	4	3	2	3	6	6	5				

Answer the following using Banker's algorithm.

- Is the system in safe state? If so, what is the safe sequence?
- If request from process P2 (0, 1, 1, 1) is considered immediately, what is the System state and Sequence? **(14 Marks)**

OR

- Which are the commonly used strategies to select a free hole from the available holes? **(06 Marks)**
 - With suitable diagram, explain external fragmentation. **(04 Marks)**
 - With neat diagram, explain paging hardware with TLB. **(10 Marks)**

Module-4

- What is Demand Paging? Explain the steps in handling page fault using appropriate diagram. **(10 Marks)**
 - Consider the page reference string : 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 for a memory with 3 frames. Determine the number of page faults using Optimal and LRU replacement algorithms. Which algorithm is most efficient? **(10 Marks)**

OR

- With neat diagram, explain Two – level and Three – level directory structure. **(08 Marks)**
 - Explain Contiguous and Linked disk space allocation methods with diagram. **(12 Marks)**

Module-5

- A drive has 200 cylinders 0 to 199. Head starts at 53 to serve the request queue : 98, 183, 37, 122, 14, 124, 65, 67. Draw disk head schedule diagram and explain for FCFS , SSTF, C – SCAN and C – LOCK. **(12 Marks)**
 - How the Access matrix model of protection can be viewed in OS? **(08 Marks)**

OR

- With neat diagram, explain SAN and MULTICS. **(08 Marks)**
 - Explain the components of a Linux System. **(06 Marks)**
 - Explain in brief fork () and exec () system calls in Linux / UNIX OS, also write a program to implement these system calls in C language. **(06 Marks)**

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
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Note : Fill in the blanks with suitable words given :

1. Nimma heSarU _____?
a) Yaaru b) eenu c) elli d) YaaVaga
2. Ee Hannina _____estu?
a) Banna b) Ruchi c) Bele d) Tuka
3. _____hudugi
a) Avalu b) Avaru c) Avanu d) Ivaru
4. Nimma KaaleJu _____?
a) Ellige b) Ellide c) Elli d) Ellinda
5. Ravi nanna _____(Friend)
a) Tamma b) Anna c) Snehita d) ShikShaka

Note: Translate English words to Kannada :

6. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
7. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
8. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa

9. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
10. Water _____.
a) HaalU b) Male c) Kere d) NeerU
11. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
12. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note: Substitute the word from the following each sentence in appropriate places:

13. Raja : Rajanaddu :: Rani : _____.
a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
14. Mane : Maneyalli :: uuru : _____.
a) eelli b) Nagardalli c) uurinalli d) Halliyalli
15. HoSa : HoSadu :: HaLeya : _____.
a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
16. oota : ootakke :: Aata : _____.
a) patakke b) AataVadu c) Aatadalli d) Aatakke
17. Mane : Manege :: KaaleJu : _____.
a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
18. Mara : Maragalu :: Hannu : _____.
a) Hannige b) Hannalli c) Henu d) Hannugalu

Note: Write the English word for the given Kannada word:

19. PradhyaPaKa _____.
a) Teacher b) Warden c) Professor d) Principal
20. Pashchima _____.
a) East b) West c) South d) North
21. Varsha _____.
a) Month b) Week c) Year d) Day
22. RaSaayanaShastra _____.
a) Chemistry b) Physics c) Botony d) Mathematics
23. HaSiru _____.
a) Red b) Blue c) Brown d) Green

24. Mara _____
a) Plant b) Earth c) Tree d) Soil
25. Granthalaya _____
a) Laboratory b) Library c) Book d) Classroom

Note: Translate following sentences into English:

26. Niivu Yarru?
a) How are you? b) Who is he?
c) Who are you? d) Where are you?
27. Ninna KaaleJu Ellide?
a) What is your name? b) Where is your house?
c) Where is your college? d) What do you do?
28. Avala heSaru Kamala.
a) Her name is Kamala. b) Kamala where are you?
c) His name is Kamala. d) Kamala good girl.
29. Idu nanna mane.
a) Where is your house? b) Which is your place?
c) This is my house. d) What is your house name?
30. Bengaluru dodda Nagara.
a) My place is Bangalore. b) Bangalore is a green city.
c) Bangalore is a city. d) Bangalore is a big city.

Note: Choose the appropriate answer:

31. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
32. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
33. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
34. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
35. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
36. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli

37. HoSadu _____(Identify the Kannada opposite word)
 a) Mele b) Kadime c) Belaku d) Haleyadu
38. Young sister _____.
 a) Tamma b) Tande c) Appa d) Tangi
39. Drama _____
 a) Kate b) Paata c) Naataka d) Banna
40. Rent _____
 a) Mane b) Bele c) Angadi d) Baadige

Note : Fill in the blanks with appropriate Kannada Word:

41. Naale _____uurige hogonaVaa? (we)
 a) Naavu b) adu c) Yaaru d) AVarU
42. Nimma _____Yaavudu? (Village)
 a) Mane b) pattana c) uuru d) byagu
43. Naanu _____bengulurige hogue? (Tomorrow)
 a) Indu b) Naadiddu c) Bega d) Naale
44. Nimma Kaaleju _____? (which)
 a) Yaake b) elli c) YaaVudu d) Hege
45. Nimma _____enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
46. Adu Namma _____ (house)
 a) Shaale b) uuru c) Pennu d) Mane

Note : Translate following Kannada sentences into English

47. Adu Enu?
 a) Who is this? b) Where is this? c) What is that? d) What is this?
48. Krishna Yaaru?
 a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
49. Niivu Yaaru?
 a) Who is she? b) Who are you? c) Who is this? d) How is?
50. AVaru Namma Tande.
 a) He is my son. b) She is my daughter.
 c) This is my place. d) He is my Father.

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Question Paper Version : B

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
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4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate following Kannada sentences into English

1. Adu Enu?
a) Who is this? b) Where is this? c) What is that? d) What is this?
2. Krishna Yaaru?
a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
3. Niivu Yaaru?
a) Who is she? b) Who are you? c) Who is this? d) How is?
4. AVaru Namma Tande.
a) He is my son. b) She is my daughter.
c) This is my place. d) He is my Father.

Note : Fill in the blanks with appropriate Kannada Word:

5. Naale _____uurige hogonaVaa? (we)
a) Naavu b) adu c) Yaaru d) AVarU
6. Nimma _____Yaavudu? (Village)
a) Mane b) pattana c) uuru d) byagu
7. Naanu _____bengulurige hogue? (Tomorrow)
a) Indu b) Naadiddu c) Bega d) Naale
8. Nimma Kaaleju _____? (which)
a) Yaake b) elli c) YaaVudu d) Hege

9. Nimma _____enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
10. Adu Namma _____ (house)
 a) Shaale b) uuru c) Pennu d) Mane

Note: Choose the appropriate answer:

11. Country _____.
 a) Rajya b) DeSha c) prapancha d) VideSha
12. Choose the mathematical '+' symbol in Kannada.
 a) Kale b) Koodu c) Gunisu d) BaagiSu
13. Daughter _____.
 a) Maga b) Akka c) Hudugi d) Magalu
14. Samskruthi _____.
 a) Festival b) Heritage c) Culture d) All of these
15. Aarogya _____.
 a) Wealth b) Health c) Patient d) Hospital
16. Them _____.
 a) Avaru b) NaaVu c) Avanu d) elli
17. HoSadu _____(Identify the Kannada opposite word)
 a) Mele b) Kadime c) Belaku d) Haleyadu
18. Young sister _____.
 a) Tamma b) Tande c) Appa d) Tangi
19. Drama _____.
 a) Kate b) Paata c) Naataka d) Banna
20. Rent _____.
 a) Mane b) Bele c) Angadi d) Baadige

Note: Translate following sentences into English:

21. Niivu Yarru?
 a) How are you? b) Who is he?
 c) Who are you? d) Where are you?
22. Ninna KaaleJu Ellide?
 a) What is your name? b) Where is your house?
 c) Where is your college? d) What do you do?
23. Avala heSaru Kamala.
 a) Her name is Kamala. b) Kamala where are you?
 c) His name is Kamala. d) Kamala good girl.

24. Idu nanna mane.
 a) Where is your house? b) Which is your place?
 c) This is my house. d) What is your house name?
25. Bengaluru dodda Nagara.
 a) My place is Bangalore. b) Bangalore is a green city.
 c) Bangalore is a city. d) Bangalore is a big city.

Note: Write the English word for the given Kannada word:

26. PradhyaPaKa _____
 a) Teacher b) Warden c) Professor d) Principal
27. Pashchima _____
 a) East b) West c) South d) North
28. Varsha _____
 a) Month b) Week c) Year d) Day
29. \RaSaayanaShastra _____
 a) Chemistry b) Physics c) Botony d) Mathematics
30. HaSiru _____
 a) Red b) Blue c) Brown d) Green
31. Mara _____
 a) Plant b) Earth c) Tree d) Soil
32. Granthalaya _____
 a) Laboratory b) Library c) Book d) Classroom

Note: Substitute the word from the following each sentence in appropriate places:

33. Raja : Rajanaddu :: Rani : _____
 a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
34. Mane : Maneyalli :: uuru : _____
 a) eelli b) Nagardalli c) uurinalli d) Halliyalli
35. HoSa : HoSadu :: HaLeya : _____
 a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
36. oota : ootakke :: Aata : _____
 a) patakke b) AataVadu c) Aatadalli d) Aatakke
37. Mane : Manege :: KaaleJu : _____
 a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
38. Mara : Maragalu :: Hannu : _____
 a) Hannige b) Hannalli c) Henu d) Hannugalu

Note: Translate English words to Kannada :

39. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
40. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
41. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa
42. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
43. Water _____.
a) HaalU b) Male c) Kere d) NeerU
44. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
45. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note : Fill in the blanks with suitable words given :

46. Nimma heSarU _____.
a) Yaaru b) eenu c) elli d) YaaVaga
47. Ee Hannina _____estu?
a) Banna b) Ruchi c) Bele d) Tuka
48. _____hudugi
a) Avalu b) Avaru c) Avanu d) Ivaru
49. Nimma KaaleJu _____.
a) Ellige b) Ellide c) Elli d) Ellinda
50. Ravi nanna _____(Friend)
a) Tamma b) Anna c) Snehita d) ShikShaka

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Question Paper Version : C

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note: Translate English words to Kannada :

1. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
2. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
3. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa
4. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
5. Water _____.
a) HaalU b) Male c) Kere d) NeerU
6. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
7. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note: Write the English word for the given Kannada word:

8. PradhyaPaKa _____.
a) Teacher b) Warden c) Professor d) Principal
9. Pashchima _____.
a) East b) West c) South d) North

10. Varsha _____
a) Month b) Week c) Year d) Day
11. RaSaayanaShastra _____
a) Chemistry b) Physics c) Botony d) Mathematics
12. HaSiru _____
a) Red b) Blue c) Brown d) Green
13. Mara _____
a) Plant b) Earth c) Tree d) Soil
14. Granthalaya _____
a) Laboratory b) Library c) Book d) Classroom

Note : Translate following Kannada sentences into English

15. Adu Enu?
a) Who is this? b) Where is this? c) What is that? d) What is this?
16. Krishna Yaaru?
a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
17. Niivu Yaaru?
a) Who is she? b) Who are you? c) Who is this? d) How is?
18. AVaru Namma Tande.
a) He is my son. b) She is my daughter.
c) This is my place. d) He is my Father.

Note: Choose the appropriate answer:

19. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
20. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
21. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
22. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
23. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
24. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli

25. HoSadu _____ (Identify the Kannada opposite word)
 a) Mele b) Kadime c) Belaku d) Haleyadu
26. Young sister _____.
 a) Tamma b) Tande c) Appa d) Tangi
27. Drama _____
 a) Kate b) Paata c) Naataka d) Banna
28. Rent _____
 a) Mane b) Bele c) Angadi d) Baadige

Note : Fill in the blanks with suitable words given :

29. Nimma heSarU _____?
 a) Yaaru b) eenu c) elli d) YaaVaga
30. Ee Hannina _____estu?
 a) Banna b) Ruchi c) Bele d) Tuka
31. _____hudugi
 a) Avalu b) Avaru c) Avanu d) Ivaru
32. Nimma KaaleJu _____?
 a) Ellige b) Ellide c) Elli d) Ellinda
33. Ravi nanna _____ (Friend)
 a) Tamma b) Anna c) Snehita d) ShikShaka

Note : Fill in the blanks with appropriate Kannada Word:

34. Naale _____uurige hogonaVaa? (we)
 a) Naavu b) adu c) Yaaru d) AVarU
35. Nimma _____Yaavudu? (Village)
 a) Mane b) pattana c) uuru d) byagu
36. Naanu _____bengulurige hogueve? (Tomorrow)
 a) Indu b) Naadiddu c) Bega d) Naale
37. Nimma Kaaleju _____? (which)
 a) Yaake b) elli c) YaaVudu d) Hege
38. Nimma _____enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
39. Adu Namma _____ (house)
 a) Shaale b) uuru c) Pennu d) Mane

Note: Translate following sentences into English:

40. Niivu Yarru?
a) How are you? b) Who is he?
c) Who are you? d) Where are you?
41. Ninna KaaleJu Ellide?
a) What is your name? b) Where is your house?
c) Where is your college? d) What do you do?
42. Avala heSaru Kamala.
a) Her name is Kamala. b) Kamala where are you?
c) His name is Kamala. d) Kamala good girl.
43. Idu nanna mane.
a) Where is your house? b) Which is your place?
c) This is my house. d) What is your house name?
44. Bengaluru dodda Nagara.
a) My place is Bangalore. b) Bangalore is a green city.
c) Bangalore is a city. d) Bangalore is a big city.

Note: Substitute the word from the following each sentence in appropriate places:

45. Raja : Rajanaddu :: Rani : _____
a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
46. Mane : Maneyalli :: uuru : _____
a) eelli b) Nagardalli c) uurinalli d) Halliyalli
47. HoSa : HoSadu :: HaLeya : _____
a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
48. oota : ootakke :: Aata : _____
a) patakke b) AataVadu c) Aatadalli d) Aatakke
49. Mane : Manege :: KaaleJu : _____
a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
50. Mara : Maragalu :: Hannu : _____
a) Hannige b) Hannalli c) Hennu d) Hannugalu

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Question Paper Version : D

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note: Translate following sentences into English:

1. Niiyu Yarru?
a) How are you? b) Who is he?
c) Who are you? d) Where are you?
2. Ninna KaaleJu Ellide?
a) What is your name? b) Where is your house?
c) Where is your college? d) What do you do?
3. Avala heSaru Kamala.
a) Her name is Kamala. b) Kamala where are you?
c) His name is Kamala. d) Kamala good girl.
4. Idu nanna mane.
a) Where is your house? b) Which is your place?
c) This is my house. d) What is your house name?
5. Bengaluru dodda Nagara.
a) My place is Bangalore. b) Bangalore is a green city.
c) Bangalore is a city. d) Bangalore is a big city.

Note: Substitute the word from the following each sentence in appropriate places:

6. Raja : Rajanaddu :: Rani : _____
a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
7. Mane : Maneyalli :: uuru : _____
a) eelli b) Nagardalli c) uurinalli d) Halliyalli

8. HoSa : HoSadu :: HaLeya : _____
 a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
9. oota : ootakke :: Aata : _____
 a) patakke b) AataVadu c) Aatadalli d) Aatakke
10. Mane : Manege :: KaaleJu : _____
 a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
11. Mara : Maragalu :: Hannu : _____
 a) Hannige b) Hannalli c) Hennu d) Hannugalu

Note : Fill in the blanks with suitable words given :

12. Nimma heSarU _____?
 a) Yaaru b) eenu c) elli d) YaaVaga
13. Ee Hannina _____estu?
 a) Banna b) Ruchi c) Bele d) Tuka
14. _____hudugi
 a) Avalu b) Avaru c) Avanu d) Ivaru
15. Nimma KaaleJu _____?
 a) Ellige b) Ellide c) Elli d) Ellinda
16. Ravi nanna _____(Friend)
 a) Tamma b) Anna c) Snehita d) ShikShaka

Note : Fill in the blanks with appropriate Kannada Word:

17. Naale _____uurige hogonaVaa? (we)
 a) Naavu b) adu c) Yaaru d) AVarU
18. Nimma _____Yaavudu? (Village)
 a) Mane b) pattana c) uuru d) byagu
19. Naanu _____bengulurige hogue? (Tomorrow)
 a) Indu b) Naadiddu c) Bega d) Naale
20. Nimma Kaaleju _____? (which)
 a) Yaake b) elli c) YaaVudu d) Hege
21. Nimma _____enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
22. Adu Namma _____ (house)
 a) Shaale b) uuru c) Pennu d) Mane

Note: Choose the appropriate answer:

23. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
24. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
25. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
26. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
27. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
28. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli
29. HoSadu _____(Identify the Kannada opposite word)
a) Mele b) Kadime c) Belaku d) Haleyadu
30. Young sister _____.
a) Tamma b) Tande c) Appa d) Tangi
31. Drama _____.
a) Kate b) Paata c) Naataka d) Banna
32. Rent _____.
a) Mane b) Bele c) Angadi d) Baadige

Note: Translate English words to Kannada :

33. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
34. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
35. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa
36. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
37. Water _____.
a) HaalU b) Male c) Kere d) NeerU

38. Child _____.
 a) Magalu b) Huduga c) Magu d) Mahile
39. Good night _____.
 a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note : Translate following Kannada sentences into English

40. Adu Enu?
 a) Who is this? b) Where is this? c) What is that? d) What is this?
41. Krishna Yaaru?
 a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
42. Niivu Yaaru?
 a) Who is she? b) Who are you? c) Who is this? d) How is?
43. AVaru Namma Tande.
 a) He is my son. b) She is my daughter.
 c) This is my place. d) He is my Father.

Note: Write the English word for the given Kannada word:

44. PradhyaPaKa _____.
 a) Teacher b) Warden c) Professor d) Principal
45. Pashchima _____.
 a) East b) West c) South d) North
46. Varsha _____.
 a) Month b) Week c) Year d) Day
47. RaSaayanaShastra _____.
 a) Chemistry b) Physics c) Botony d) Mathematics
48. HaSiru _____.
 a) Red b) Blue c) Brown d) Green
49. Mara _____.
 a) Plant b) Earth c) Tree d) Soil
50. Granthalaya _____.
 a) Laboratory b) Library c) Book d) Classroom

* * * * *

1. ಮಂಕುಕುಮಿನ ಕೆ ಕೃಷಿಯನ್ನು ಬಿಡಿದವರು.

- a) ಕುಂಟೆಂಪಿ b) ದ. ತಿ. ರ. ಹಿಂಪಿ, c) ಕನಕಾಸ d) 8. 9. ಕ

2. ವಾಕುತಂ ಕವನ ಸಂಗ್ರಹವು ----- ಪರಿಚಿತವಾಗಿದೆ.

- a) ಪದ್ಮಾಭರಣ b) ಪದ್ಮಾಭರಣ c) ಕಾಂನಾಂರ d) ಪದ್ಮಾಂ

37. ಕಾಂನಾಂರ ಪದ್ಯ ಅಥವಾ

- a) ಕಾಂನಾಂರ b) ಅತಿ c) ಕಾಂನಾಂರ d) ಹಣ

38. ಕನಕಾಸವು ಒಂದು ಕಾಂನಾಂರ ಪದ್ಯವನ್ನು ತಂದುಕೊಟ್ಟವರು

- a) 8. 9. ಕ b) ಕುಂಟೆಂಪಿ c) ದ. ತಿ. ರ. ಹಿಂಪಿ, d) ಕನಕಾಸ

39. ಕುರುಡು ಕಾಂನಾಂರವನ್ನು ಕಾವ್ಯ ಕವನಸಂಕಲನವೆಂದು ಆಯ್ಕೆ ಮಾಡಲಾಗಿದೆ

- a) ವಾಕುತಂ b) ವಾಕುತಂ c) ಮಂಕುತಮಿನ ಕೆ d) ವಾದಕಾಂನಾಂರ

40. 8. 9. ಕಯವರ ಕಾವ್ಯ, ವಾಕುತಂ

- a) ಮಂಕುಕುಮಿ b) ಕೂಡಲಸಂಗಮವಿವ
c) ಕನಕಾಸ d) ಕಾಂನಾಂರ

41. ಆಯ್ಕೆ 2ನೇಯ, ನವರ ವಚನದ ಕಾವ್ಯವು ಅತಿ, ಅಂತ 2ನೇಯ ಮಹತ್ವವನ್ನು ಗಳಿಸಿದೆ.

- a) ಬಡತನ b) ಲಿಪಿ c) ಬಂಗಾರ d) ವಜ್ರ

42. ಗುರು, ಕಾಂನಾಂರ ಜಂಗಮ ಈ ಮೂರನೇಯ ಅತಿ, ಅಂತ 2ನೇಯ ಮಹತ್ವವನ್ನು ಗಳಿಸಿದೆ.

- a) ಕಾಂನಾಂರ b) 2ನೇಯ c) ನಾಂನಾಂರ d) ಕಾಂನಾಂರ 2ನೇಯ

VTU-16-10-2023 09:28:40am

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Harmony should be maintained in

a) Between body and life	b) Between self and society
c) Between life and environment	d) All of these
 2. Basic human aspirations are

a) Continuous Happiness	b) Prosperity
c) Both a and b	d) None of these
 3. Purpose of value education is

a) Foster universal core values	b) Make the syllabus easy
c) Develop values in individuals	d) All of these
 4. Knowing means having the

a) Self exploration	b) Self evaluation
c) Right understanding	d) Having knowledge of all
 5. Happiness means

a) To be in a state of harmony	
b) Fulfillment of desired expectations from others	
c) Fulfillment of desired feelings from others	
d) Fulfillment of desired sensations from body	
 6. _____ is called foundation value

a) Respect	b) Affection	c) Love	d) Trust
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 7. The Human goal at the level of nature is

a) Prosperity	b) Fearlessness
c) Co-existence	d) Right understanding
 8. The outcome of justice is _____

a) Right understanding	b) Prosperity
c) Trust and Fearlessness	d) Coexistence with nature

9. Society means
 - a) Family
 - b) All human beings
 - c) Few individuals
 - d) None of these
10. Feeling for those who have made effort for excellence is _____
 - a) Excellence
 - b) Reverence
 - c) Glory
 - d) None of these
11. An individual people aspiring for the universal human order will be
 - a) More responsible socially and ecologically
 - b) More rich
 - c) More powerful
 - d) More well travel
12. A harmonious world is created by values at 4 levels there are
 - a) Home, Family, Society, Universe
 - b) Individual, Family, Society, Universe
 - c) School, Home, Office, Temple
 - d) None of these
13. "What is the innateness of Tulsi-Plant (or any plant)?
 - a) Existence + Growth
 - b) Cruelty/non-cruelty
 - c) Nurture/Worsen
 - d) Will to live in self
14. As individual people anticipates for the universal human order may be
 - a) More rich
 - b) More Powerful
 - c) More responsible socially and ecologically
 - d) More well-travelled
15. Understanding of human values leads to the practice of
 - a) Responsibilities
 - b) Ethics
 - c) Profession
 - d) Professional ethics
16. "Seeing the self by the self" means
 - a) The consciousness observing the consciousness
 - b) The consciousness observing the material
 - c) The consciousness observing the co-existence
 - d) None of these
17. Education-Right living leads to _____
 - a) Right understanding
 - b) Confusions
 - c) Doubts
 - d) None of these
18. There are _____ dimensions of living
 - a) Six (6)
 - b) Four (4)
 - c) Nine (9)
 - d) Five (5)
19. The second order of nature is _____
 - a) Material order
 - b) Bio order
 - c) Animal order
 - d) Human order
20. The innateness of material order is _____
 - a) Existence
 - b) Cruelty/Non-cruelty
 - c) Growth
 - d) Nurture/Worsen
21. Comprehensive human goal at the level of individual
 - a) Prosperity
 - b) Fearlessness
 - c) Co-existence
 - d) Right understanding
22. The fourth order of nature is
 - a) Material
 - b) Bio order
 - c) Animal order
 - d) Human order

23. _____ is the basic unit of human society.
 a) Group b) Individual c) Nature d) Society
24. Which of the following is a positive value?
 a) Self respect b) Anger
 c) Fear d) Narrow mindedness
25. What is the nature of self?
 a) Conscious b) Physio-Chemical
 c) Biochemical d) Semi-Conscious
26. Self exploration is seeing beyond
 a) Universe b) House c) Box d) Your senses
27. Human to human interaction is called as
 a) Work b) Behaviour c) System d) None of these
28. “All are our own, all are interconnected, interdependent” means.
 a) Oneness b) Worship c) Ease d) None of these
29. Prosperity means
 a) Feeling of having more than required physical facility
 b) Feeling of having less physical facility
 c) Feeling of having more physical facility
 d) Deprived of physical facility
30. Realization is the activity of
 a) Self b) Body
 c) Both self and Body d) None of these
31. Who is responsible for happiness and unhappiness?
 a) Self b) Body
 c) Outside situation d) Society
32. Value education helps to
 a) Removes our contradictions
 b) Remove our confusions
 c) Bring harmony at all levels of human living
 d) All of these
33. Undivided society is ensured by
 a) Right understanding b) Physical facilities
 c) Relationship with right understanding d) None of these
34. The third order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
35. Which is the complete value?
 a) Love b) Respect c) Trust d) Care
36. Body is a _____ unit while the self is a _____ unit.
 a) Material, Consciousness b) Consciousness, Material
 c) Material, Material d) Consciousness, Consciousness

37. Society is an extension of _____
 a) Human Being b) Family c) Nature d) Existence
38. In value education *Sanyam* means _____
 a) Self-exploration b) Self-evolution
 c) Self-regulation d) None of these
39. To fulfill human aspirations _____ are necessary
 a) Both values and skills b) Values
 c) Skills d) None of these
40. The innateness of Bio order is _____
 a) Existence b) Cruelty / Non-cruelty
 c) Growth d) Nurture / Worsen
41. The feelings for those who have made effort for my excellence is _____
 a) Excellence b) Reverence c) Gratitude d) Glory
42. There are _____ comprehensive human goals
 a) Eight (8) b) Six (6) c) Four (4) d) Nine (9)
43. The first order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
44. Developing ethical competence in the profession is the only effective way to ensure
 a) Responsibilities b) Ethics
 c) Profession d) Professional ethics
45. Ensuring justice in relationship, on the basis of values leads to _____ in society.
 a) Fearlessness b) Trust
 c) Fearlessness and Trust d) None of these
46. Human goal → match correctly
 A Right understanding 1 Individual
 B Prosperity 2 Family
 C Fearlessness (Trust) 3 Society
 D Co-existence 4 Nature/Existence
 a) 1 – D, 2 – A, 3 – B, 4 – C b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4 d) 1 – C, 2 – D, 3 – A, 4 – B
47. The human goal at the level of family is
 a) Prosperity b) Fearlessness
 c) Co-existence d) Right understanding
48. *Sah-astitua* means _____
 a) Co-existence b) Co-operation
 c) Co-option d) Corporate identity
49. Acceptance of excellence in others is called
 a) Reverence b) Guidance
 c) Glory d) Gratitude
50. Each human being is co-existence of
 a) Spirit and *Sanyam* b) Health and prosperity
 c) Self and Body d) Mind and Soul

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Question Paper Version : B

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023 Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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-
1. The feelings for those who have made effort for my excellence is _____
 a) Excellence b) Reverence c) Gratitude d) Glory
 2. There are _____ comprehensive human goals
 a) Eight (8) b) Six (6) c) Four (4) d) Nine (9)
 3. The first order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
 4. Developing ethical competence in the profession is the only effective way to ensure
 a) Responsibilities b) Ethics
 c) Profession d) Professional ethics
 5. Ensuring justice in relationship, on the basis of values leads to _____ in society.
 a) Fearlessness b) Trust
 c) Fearlessness and Trust d) None of these
 6. Human goal → match correctly

A Right understanding	1 Individual
B Prosperity	2 Family
C Fearlessness (Trust)	3 Society
D Co-existence	4 Nature/Existence

 a) 1 – D, 2 – A, 3 – B, 4 – C b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4 d) 1 – C, 2 – D, 3 – A, 4 – B
 7. The human goal at the level of family is
 a) Prosperity b) Fearlessness
 c) Co-existence d) Right understanding
 8. Sah-astitua means _____
 a) Co-existence b) Co-operation
 c) Co-option d) Corporate identity

9. Acceptance of excellence in others is called
 - a) Reverence
 - b) Guidance
 - c) Glory
 - d) Gratitude
10. Each human being is co-existence of
 - a) Spirit and Sanyam
 - b) Health and prosperity
 - c) Self and Body
 - d) Mind and Soul
11. Who is responsible for happiness and unhappiness?
 - a) Self
 - b) Body
 - c) Outside situation
 - d) Society
12. Value education helps to
 - a) Removes our contradictions
 - b) Remove our confusions
 - c) Bring harmony at all levels of human living
 - d) All of these
13. Undivided society is ensured by
 - a) Right understanding
 - b) Physical facilities
 - c) Relationship with right understanding
 - d) None of these
14. The third order of nature is _____
 - a) Material order
 - b) Bio order
 - c) Animal order
 - d) Human order
15. Which is the complete value?
 - a) Love
 - b) Respect
 - c) Trust
 - d) Care
16. Body is a _____ unit while the self is a _____ unit.
 - a) Material, Consciousness
 - b) Consciousness, Material
 - c) Material, Material
 - d) Consciousness, Consciousness
17. Society is an extension of _____
 - a) Human Being
 - b) Family
 - c) Nature
 - d) Existence
18. In value education Sanyam means _____
 - a) Self-exploration
 - b) Self-evolution
 - c) Self-regulation
 - d) None of these
19. To fulfill human aspirations _____ are necessary
 - a) Both values and skills
 - b) Values
 - c) Skills
 - d) None of these
20. The innateness of Bio order is _____
 - a) Existence
 - b) Cruelty / Non-cruelty
 - c) Growth
 - d) Nurture / Worsen
21. An individual people aspiring for the universal human order will be
 - a) More responsible socially and ecologically
 - b) More rich
 - c) More powerful
 - d) More well travel
22. A harmonious world is created by values at 4 levels there are
 - a) Home, Family, Society, Universe
 - b) Individual, Family, Society, Universe
 - c) School, Home, Office, Temple
 - d) None of these

23. "What is the innateness of Tulsi-Plant (or any plant)?

 - a) Existence + Growth
 - b) Cruelty/non-cruelty
 - c) Nurture/Worsen
 - d) Will to live in self
24. As individual people anticipates for the universal human order may be

 - a) More rich
 - b) More Powerful
 - c) More responsible socially and ecologically
 - d) More well-travelled
25. Understanding of human values leads to the practice of

 - a) Responsibilities
 - b) Ethics
 - c) Profession
 - d) Professional ethics
26. "Seeing the self by the self" means

 - a) The consciousness observing the consciousness
 - b) The consciousness observing the material
 - c) The consciousness observing the co-existence
 - d) None of these
27. Education-Right living leads to _____

 - a) Right understanding
 - b) Confusions
 - c) Doubts
 - d) None of these
28. There are _____ dimensions of living

 - a) Six (6)
 - b) Four (4)
 - c) Nine (9)
 - d) Five (5)
29. The second order of nature is _____

 - a) Material order
 - b) Bio order
 - c) Animal order
 - d) Human order
30. The innateness of material order is _____

 - a) Existence
 - b) Cruelty/Non-cruelty
 - c) Growth
 - d) Nurture/Worsen
31. Harmony should be maintained in

 - a) Between body and life
 - b) Between self and society
 - c) Between life and environment
 - d) All of these
32. Basic human aspirations are

 - a) Continuous Happiness
 - b) Prosperity
 - c) Both a and b
 - d) None of these
33. Purpose of value education is

 - a) Foster universal core values
 - b) Make the syllabus easy
 - c) Develop values in individuals
 - d) All of these
34. Knowing means having the

 - a) Self exploration
 - b) Self evaluation
 - c) Right understanding
 - d) Having knowledge of all
35. Happiness means

 - a) To be in a state of harmony
 - b) Fulfillment of desired expectations from others
 - c) Fulfillment of desired feelings from others
 - d) Fulfillment of desired sensations from body

36. _____ is called foundation value
a) Respect b) Affection c) Love d) Trust
37. The Human goal at the level of nature is
a) Prosperity b) Fearlessness
c) Co-existence d) Right understanding
38. The outcome of justice is _____
a) Right understanding b) Prosperity
c) Trust and Fearlessness d) Coexistence with nature
39. Society means
a) Family b) All human beings
c) Few individuals d) None of these
40. Feeling for those who have made effort for excellence is _____
a) Excellence b) Reverence
c) Glory d) None of these
41. Comprehensive human goal at the level of individual
a) Prosperity b) Fearlessness
c) Co-existence d) Right understanding
42. The fourth order of nature is
a) Material b) Bio order c) Animal order d) Human order
43. _____ is the basic unit of human society.
a) Group b) Individual c) Nature d) Society
44. Which of the following is a positive value?
a) Self respect b) Anger
c) Fear d) Narrow mindedness
45. What is the nature of self?
a) Conscious b) Physio-Chemical
c) Biochemical d) Semi-Conscious
46. Self exploration is seeing beyond
a) Universe b) House c) Box d) Your senses
47. Human to human interaction is called as
a) Work b) Behaviour c) System d) None of these
48. “All are our own, all are interconnected, interdependent” means.
a) Oneness b) Worship c) Ease d) None of these
49. Prosperity means
a) Feeling of having more than required physical facility
b) Feeling of having less physical facility
c) Feeling of having more physical facility
d) Deprived of physical facility
50. Realization is the activity of
a) Self b) Body
c) Both self and Body d) None of these

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Universal Human Values

Time: 1 hr.]

[Max. Marks: 50]

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- Who is responsible for happiness and unhappiness?
 - Self
 - Body
 - Outside situation
 - Society
- Value education helps to
 - Removes our contradictions
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 - Bring harmony at all levels of human living
 - All of these
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- The third order of nature is _____
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27. The human goal at the level of family is
 a) Prosperity b) Fearlessness
 c) Co-existence d) Right understanding
28. Sah-astitua means _____
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 c) Co-option d) Corporate identity
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49. The second order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
50. The innateness of material order is _____
 a) Existence b) Cruelty/Non-cruelty
 c) Growth d) Nurture/Worsen

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Question Paper Version : D

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023 Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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-
1. Comprehensive human goal at the level of individual

a) Prosperity	b) Fearlessness
c) Co-existence	d) Right understanding
 2. The fourth order of nature is

a) Material	b) Bio order	c) Animal order	d) Human order
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 3. _____ is the basic unit of human society.

a) Group	b) Individual	c) Nature	d) Society
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c) Fear	d) Narrow mindedness
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c) Biochemical	d) Semi-Conscious
 6. Self exploration is seeing beyond

a) Universe	b) House	c) Box	d) Your senses
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 7. Human to human interaction is called as

a) Work	b) Behaviour	c) System	d) None of these
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 8. “All are our own, all are interconnected, interdependent” means.

a) Oneness	b) Worship	c) Ease	d) None of these
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 9. Prosperity means

a) Feeling of having more than required physical facility
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 c) Fearlessness and Trust
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16. Human goal → match correctly
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 C Fearlessness (Trust) 3 Society
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 a) 1 – D, 2 – A, 3 – B, 4 – C b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4 d) 1 – C, 2 – D, 3 – A, 4 – B
17. The human goal at the level of family is
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18. Sah-astitua means _____
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19. Acceptance of excellence in others is called
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20. Each human being is co-existence of
 a) Spirit and Sanyam
 b) Health and prosperity
 c) Self and Body
 d) Mind and Soul
21. Who is responsible for happiness and unhappiness?
 a) Self
 b) Body
 c) Outside situation
 d) Society
22. Value education helps to
 a) Removes our contradictions
 b) Remove our confusions
 c) Bring harmony at all levels of human living
 d) All of these

Version – D – 4 of 4

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18CS61

Sixth Semester B.E. Degree Examination, June/July 2023

System Software and Compilers

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With reference to SIC/XE machine architectures explain instruction formats and addressing modes, clearly indicating the settings of different flag bits. (10 Marks)
- b. With an illustrate example, explain the need for a two pass assembler. Explain the data structures used in 2-pass assembler. Mention their functions clearly during pass 1 and pass 2. (10 Marks)

OR

- 2 a. Generate the complete object program for the following SIC/XE assembly language programs. Assume : CLEAR = B4, LDT = 74, TD = EO, JEQ = 30, TIXR = B8, JLT = 38, RSUB = 4C, LDCH = 50, WD = DC, X = 1, T = 5.

```

WRREC  START      105D
        CLEAR      X
        LDT         LENGTH
WLOOP   TD         OUTPUT
        JEQ         WLOOP
        LDCH        BUFFER, X
        WD          OUTPUT
        TIXR        T
        JLT         WLOOP
        RSUB
OUTPUT  BYTE       X'O5'
BUFFER  RESB       400
LENGTH  RESB       2
        END        WRREC
    
```

- b. Explain the absolute loader and Bootstrap loader with algorithm/source code. (10 Marks)

Module-2

- 3 a. What is a Compiler? Explain the various phases of a compiler with a neat diagram and show the output of each phase for the expression $a = b + c * 25$. Assume all variable are a type float. (10 Marks)
- b. Write a note on the commonly used compiler – construction tools. (04 Marks)
- c. Describe Input Buffering mechanism with an algorithm for lookahead code with sentinels. (06 Marks)

OR

- 4 a. Construct the transition diagrams to recognize the tokens given below and explain the same.
 - i) relop ii) Identifier iii) unsigned numbers (10 Marks)
- b. With example, define the operations on languages. (04 Marks)
- c. Discuss the issues/errors of lexical analysis and the error recovery actions that can be performed. (06 Marks)

Module-3

- 5 a. What is recursive-decent parsing? Explain with a pseudocode. Take the grammar $S \rightarrow cAd$, $A \rightarrow ab|a$ as an example and trace it for input string $w = cad$. Explain how backtracking can be used for tracing. (10 Marks)
- b. Consider the context free grammar :
 $S \rightarrow SS + | SS * | a$ and string $w = aa + a^*$
- Give the leftmost and rightmost derivation and parse tree for the string
 - Is the grammar ambiguous or unambiguous? Justify your answer
 - Eliminate left Recursion
- (10 Marks)

OR

- 6 a. With a neat diagram, explain the model of a table driven predictive parser. Write and explain the predictive parsing algorithm. (10 Marks)
- b. Consider the following grammar with terminals $(, [,),]$.
 $S \rightarrow TS | [S] S |)S | \epsilon$
 $T \rightarrow (X)$
 $X \rightarrow TX | [X] X | \epsilon$
- Construct FIRST and FOLLOW sets
 - Construct its LL(1) parsing table
 - Is this grammar LL(1)?
- (10 Marks)

Module-4

- Explain the meta – characters used in regular expression with examples. (10 Marks)
- 7 a. Write a LEX program to recognize and count the number of identifiers in a given input file. (10 Marks)
- b. Show how the program is compiled and executed. (10 Marks)

OR

- 8 a. What are the ambiguities that arise while evaluating a regular expression? Explain with example. (10 Marks)
- b. Write a YACC program to recognize a valid arithmetic expression that uses operators $+, -, *$ and $/$. (10 Marks)

Module-5

- 9 a. What is a dependency graph? Give a syntax directed definition for simple type declaration including int and float type. Construct annotated parse tree and dependency graph for the input, float a, b, c. (10 Marks)
- b. Explain synthesized attribute, inherited attribute, S – attributed definition and L- attributed definitions with examples. (10 Marks)

OR

- 10 a. What is a three – address code? explain the different ways of representing three – address codes with examples. (10 Marks)
- b. What is target computer model? Explain the different kinds of instructions and addressing modes available in assembly language or a target machine. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023 Computer Graphics and Visualization

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is computer graphics? Mention the list of applications. How they are classified? (06 Marks)
b. Explain with neat diagram operation of cathode-Ray tubes and shadow-mask CRT. (08 Marks)
c. Explain the logical organization of the video-controller. (06 Marks)

OR

- 2 a. Write Bresenham's line drawing algorithm. Using Bresenham's algorithm calculate the pixel positions for the screen coordinates (1, 1) and (6, 7). (10 Marks)
b. Write midpoint circle algorithm. Draw the circle with 8 as radius. (10 Marks)

Module-2

- 3 a. Explain scanline polygon filling algorithm with neat sketches and example. (06 Marks)
b. With a neat figure explain various polygon types in OpenGL. (06 Marks)
c. What is concatenation of transformation? Explain the following considered 2D:
i) Rotation about a fixed point
ii) Scaling about a fixed point. (08 Marks)

OR

- 4 a. Define the following two dimensional transformations translation, rotation, scaling reflection and shearing. Give example for each. (10 Marks)
b. With a neat figure explain two dimensional viewing pipeline? Explain OpenGL 2D viewing functions. (10 Marks)

Module-3

- 5 a. Explain window to view port coordinate transformation. (04 Marks)
b. Explain the Cohen Sutherland line clipping algorithm considering all cases. (08 Marks)
c. With an example explain Sutherland Hodgeman polygon clipping algorithm. (08 Marks)

OR

- 6 a. Discuss the OpenGL functions for the following 3D dimensional transformations:
i) Translation ii) Scaling iii) Rotation. (06 Marks)
b. Explain the following color models:
i) RGB color model ii) CMY color model. (08 Marks)
c. Explain the basic illumination models. (06 Marks)

Module-4

- 7 a. What is three dimensional viewing? Explain three dimensional viewing pipeline with neat diagram. (08 Marks)
- b. Explain OpenGL three dimensional viewing functions, with example for each:
i) gluLookAt ii) glOrtho iii) glPerspective iv) glFrustum. (12 Marks)

OR

- 8 a. Explain classification of visible surface detection and back face detection algorithm. (08 Marks)
- b. Explain Z-buffer or depth buffer algorithm for visible surface detection. (06 Marks)
- c. Discuss OpenGL visibility-detection functions with an example. (06 Marks)

Module-5

- 9 a. List and explain the various classes of logical input devices that are supported by OpenGL. With suitable diagrams, explain various input modes. (10 Marks)
- b. Explain how keyboard, window and mouse events are recognized by GLUT. Give suitable example. (10 Marks)

OR

- 10 a. How pop-up menus are created using GLUT? Illustrate with an example. (06 Marks)
- b. What are the features of a good interactive program? What are the advantages of double buffering? Explain. (08 Marks)
- c. Explain Bezier cubic curves. Give the properties of Bezier curves. (06 Marks)

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18CS63

Sixth Semester B.E. Degree Examination, June/July 2023

Web Technology and its Application

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the structure of HTML document with an example. (10 Marks)
- b. Explain ordered and unordered list in HTML5 with code. (10 Marks)

OR

- 2 a. Explain embedded style sheet and external style sheet with code. (10 Marks)
- b. Discuss class selector and id selector with code. (10 Marks)

Module-2

- 3 a. List and describe different button controls. (08 Marks)
- b. Explain the elements used to define the structure of HTML table. Give an example of table. (12 Marks)

OR

- 4 a. What is normal flow in the context of CSS? (10 Marks)
- b. What is responsive design? What are its four components, explain them. (10 Marks)

Module-3

- 5 a. With diagrams, explain client script and server script execution. (10 Marks)
- b. Explain the php module in the apache and describe the difference between multi-threaded and multi-process setup. (10 Marks)

OR

- 6 a. Describe different comparison operators in javascript. (12 Marks)
- b. What is DOM? Briefly explain the different types of nodes. (08 Marks)

Module-4

- 7 a. Describe polymorphism and data encapsulation in php. (10 Marks)
- b. Highlight the techniques for reading and writing of files in php with example. (10 Marks)

OR

- 8 a. What are errors and exceptions? (10 Marks)
- b. Discuss \$ - SERVER array. How is it used? (10 Marks)

Module-5

- 9 a. What are cookies? Explain php mechanism for writing and reading cookies with example. (10 Marks)
- b. Discuss strategies to caching web applications. (10 Marks)

OR

- 10 a. What is AJAX? Explain how asynchronous request is handled using UML diagrams. (10 Marks)
- b. Describe how XML is processed in javascript and php. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8=50, will be treated as malpractice.

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Sixth Semester B.E. Degree Examination, June/July 2023 Data Mining and Data Warehousing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Datawarehouse. Explain Multitier Architecture of Data Warehousing with diagram. (10 Marks)
- b. Explain ETL process with the neat diagram. (10 Marks)

OR

- 2 a. Explain the schemas of multi-dimensional data models. (10 Marks)
- b. Explain data cube operations with example. (10 Marks)

Module-2

- 3 a. Explain different methods of indexing OLAP data. (10 Marks)
- b. Explain the following preprocessing techniques : (10 Marks)

(i) Feature subset selection.
(ii) Sampling.

OR

- 4 a. List the different types of Dataset and explain with an example. (10 Marks)
- b. Consider $X = (0, 1, 0, 1)$, $Y = (1, 0, 1, 0)$. Find cosine, correlation, Euclidean, Jaccard and SMC. (10 Marks)

Module-3

- 5 a. A database has five transactions. Let min sup = 60% and min conf = 80%.

Table Q5 (a)

TID	Items.bought
T ₁₀₀	{M,O,N,K,E,Y}
T ₂₀₀	{D,O,N,K,E,Y}
T ₃₀₀	{M,A,K,E}
T ₄₀₀	{M,U,C,K,Y}
T ₅₀₀	{C,O,O,K,I,E}

- (i) Find all frequent itemsets using Apriori Algorithm.
 - (ii) List all the strong association Rules. (12 Marks)
- b. Identify and explain the alternative methods for generating frequent itemsets. (08 Marks)

OR

- 6 a. Construct FP tree by showing tree separately after reading each transaction and find the frequent itemset generation. Consider the transaction dataset :

Table : Q6 (a)

TID	Items.bought
100	{f, a, c, d, g, i, m, p}
200	{a, b, c, f, l, m, o}
300	{b, f, h, j, o}
400	{b, c, k, s, p}
500	{a, f, c, e, l, p, m, n }

- Let min-support = 3 (12 Marks)
 - b. Explain the various measures of evaluating association patterns. (08 Marks)

Module-4

- 7 a. Explain the general approach for solving classification problem. (08 Marks)
 b. Build a decision tree using Hun't Algorithm for the given dataset.

Tid	Home owner	Marital status	Annual Income	Defaulted Borrower
1	yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Married	120 K	No
5	No	Divorced	95 K	Yes
6	No	Married	60 K	No
7	yes	Divorced	220 K	No
8	No	Single	85 K	Yes
9	No	Married	75 K	No
10	No	Single	90 K	Yes

Table Q7 (b)

(12 Marks)

OR

- 8 a. Explain K-nearest neighbor classification algorithm with example. (08 Marks)
 b. State Bays theorem and explain how bayes theorem is used in the Naïve Bayesian classifier with example. (12 Marks)

Module-5

- 9 a. Calculate the cluster for the following 8 points (x, y). Represents into 3 clusters
 $A_1(2, 10)$ $A_2(2, 5)$ $A_3(8, 4)$ $B_1(5, 8)$ $B_2(7, 5)$ $B_3(6, 4)$ $C_1(1, 2)$ $C_2(4, 9)$ (10 Marks)
 b. Explain the following :
 (i) Density based clustering.
 (ii) Graph based clustering. (10 Marks)

OR

- 10 a. What are the basic approaches used for generating a agglomerative hierarchical clustering? (10 Marks)
 b. Explain DBSCAN algorithm with example. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023

Mobile Application Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Android? With a neat diagram, explain the components of android stack. (10 Marks)
- b. What are the challenges of android applications development? (03 Marks)
- c. Describe the development process of android applications. (07 Marks)

OR

- 2 a. List and explain the android studio window panes. (08 Marks)
- b. What are the intents? Explain the different types of intents. (06 Marks)
- c. With a neat diagram, explain the activity states and the callback methods that occur or the activity transitions between different states. (06 Marks)

Module-2

- 3 a. What are the android input controls? Explain any two user input control in details. (10 Marks)
- b. What is Menu? List and explain the different types of menus. (10 Marks)

OR

- 4 a. What is a drawable? How to retrieve drawables using APIs and what are the attributes has been used to apply drawable in XML resource? List the different types of drawables and explain any one brief. (10 Marks)
- b. What are approaches to creating icons of images in Android? Define a style? How to create a style, and explain the elements of style? (10 Marks)

Module-3

- 5 a. What is the use of AsyncTask class? When AsyncTask is executed. What are the steps it goes? Explain the different kinds of AsyncTask parameters. (10 Marks)
- b. What is a service? List and explain any one service briefly. (10 Marks)

OR

- 6 a. What is a notification? How to create a notification? What are the notification design guidelines? (10 Marks)
- b. List and explain the two general types of alarms in android. (10 Marks)

Module-4

- 7 a. How do you create, save, restore shared preferences? List and explain the different modes available in shared preferences? (10 Marks)
- b. Write short notes on :
 - i) SQLite
 - ii) Transactions
 - iii) Threads
 - iv) Cursors
 - v) SQL Databases.

(10 Marks)

OR

- 8 a. What is a content provider? With a neat diagram, explain application architecture with a content provider. (10 Marks)
- b. What are loader? List the characteristics of loaders in android? What are the different classes and interfaces that may involved when using loaders in are application. (10 Marks)

Module-5

- 9 a. List and explain the different types of permissions. (04 Marks)
- b. What is Firebase? Explain firebase Real time Database with an example. (10 Marks)
- c. List and explain the different types of monitors that measures the performance of an android application. (06 Marks)

OR

- 10 a. Write short notes on :
i) Fire base analytics
ii) Fire base notifications
iii) Firebase Real time database
iv) AdMob. (10 Marks)
- b. What is an APK? What are the things required to do before generating APK in Android studio? How do you publish android application in the Google play store? (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023
Programming in Java

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the three OOP principles. (06 Marks)
- b. List and explain Control statements syntax with example. (08 Marks)
- c. Explain the following (06 Marks)
 - i) Identifiers
 - ii) Literals
 - iii) Comments.

OR

- 2 a. List and explain different Integer types. (08 Marks)
- b. Explain Java's Automatic Conversions and Casting Incompatible type. (06 Marks)
- c. Explain Initialization and Retrieval of content of different arrays in Java. (06 Marks)

Module-2

- 3 a. Explain the basic Arithmetic Operators. (08 Marks)
- b. Explain ? Operator, with example. (06 Marks)
- c. List and explain Relational Operators. (06 Marks)

OR

- 4 a. Explain the Control statements : i) if – else ii) Switch with example. (06 Marks)
- b. Explain following Jump statements : i) Continue ii) Break. (08 Marks)
- Provide proper output interpretation. (06 Marks)
- c. Explain all bitwise operators. Consider a = 3 and b = 6. Apply Bitwise logical operations on these operands and print suitable results. (06 Marks)

Module-3

- 5 a. Explain following keywords supported in Java : (08 Marks)
 - i) this ii) super iii) final iv) static.
- b. Explain Recursion in Java with example. (06 Marks)
- c. Explain different Access specifiers with example. (06 Marks)

OR

- 6 a. What is Dynamic method dispatch? Explain. (06 Marks)
- b. Explain Creation of Multilevel hierarchy in Java with example. (08 Marks)
- c. Differentiate method overloading and method overriding with example. (06 Marks)

Module-4

- 7 a. Define Interface. Explain how to define implement and assign variable in interface to perform "One interface multiple method". (06 Marks)
- b. Write a short notes on : (08 Marks)
 - i) Importing packages
 - ii) Access protection with respect to packages.
- c. Explain Exception Handling Fundamentals in Java. (06 Marks)

OR

- 8 a. Explain Uncaught Exception. (06 Marks)
b. Explain Java's checked and Unchecked exception. (08 Marks)
c. Explain Chained exception. (06 Marks)

Module-5

- 9 a. List and explain different types Wrapper with syntax. (06 Marks)
b. Explain the Fundamentals of Applets. (08 Marks)
c. Explain String Constructors of Java. (06 Marks)

OR

- 10 a. Explain Character Extraction methods of Java. (08 Marks)
b. Explain the String Modify methods : Concat () , replace () , trim () with example. (06 Marks)
c. Explain the following with respect to string buffers :
i) Append () ii) insert (). (06 Marks)

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18CS81

Eighth Semester B.E. Degree Examination, June/July 2023 Internet of Things

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define IOT. Explain in detail IOT and digitization. (06 Marks)
- b. Explain in detail with any two example IOT impact. (08 Marks)
- c. Explain the different evolutionary phases of the Internet. (06 Marks)

OR

- 2 a. Explain different challenges of IOT. (04 Marks)
- b. Explain in detail IOT World Forum (IOTWF) Standard Architecture. (08 Marks)
- c. Explain expanded view of the simplified IOT Architecture. (08 Marks)

Module-2

- 3 a. List and explain different types of sensors (any 8) with an example each. (08 Marks)
- b. What are smart objects? With neat diagram, explain characteristics of smart object. (08 Marks)
- c. What are Actuators? Explain comparison of sensors and actuators functionality with human. (04 Marks)

OR

- 4 a. What is SANET? Explain some advantages and disadvantages that a wireless based solution offers. (06 Marks)
- b. Briefly explain protocol stack utilization IEEE 802.15.4. (08 Marks)
- c. List and explain in brief communication criteria. (06 Marks)

Module-3

- 5 a. Explain key advantages of Internet Protocol for the IOT. (08 Marks)
- b. Explain the need for optimization. (08 Marks)
- c. With a neat diagram, explain comparison of an IOT protocol stack utilizing 6LOWPAN and IP protocol stack. (04 Marks)

OR

- 6 a. Write notes on Supervisory Control and Data Acquisition (SCADA). (06 Marks)
- b. Explain with neat diagram Constrained Application Protocol (COAP) message format. (08 Marks)
- c. Explain in detail Message Queuing Telemetry Transport (MQTT) publish/subscribe frame. (06 Marks)

Module-4

- 7 a. Compare: (i) Structured versus unstructured data (ii) Data in motion versus data at rest (06 Marks)
- b. With neat diagram, explain Hadoop distributed cluster and writing a file to HDFS. (08 Marks)
- c. Explain Lambda Architecture. (06 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

OR

- 8** a. Explain edge streaming analytics and functions of Edge Analytics Processing Unit. (10 Marks)
b. Explain in detail formal risk analysis structures. (10 Marks)

Module-5

- 9** a. Write notes on: (i) Arduino UNO (ii) Raspberry Pi (12 Marks)
b. With a neat diagram, explain wireless temperature monitoring system using Raspberry Pi. (08 Marks)

OR

- 10** a. Explain IOT strategy for smarter cities. (10 Marks)
b. With neat smart cities Layered Architecture diagram, explain Smart City IOT Architecture. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023

Introduction to Operating System

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define operating system. Explain the systems and users view of an operating system. (10 Marks)
- b. Explain the essential properties of batch, real time and distributed operating systems. (10 Marks)

OR

- 2 a. List and explain services provided by an operating system that are designed to make using computer systems more convenient for users. (10 Marks)
- b. With a neat diagram of VM-Ware architecture, explain the concept of virtual machine and the main advantage of using VM architecture. (10 Marks)

Module-2

- 3 a. What is process control block? List the information stored in process control block. Explain the different states of a process during its life cycle. (10 Marks)
- b. Explain shared memory and message passing technique for Inter-process communication between co-operating processes. (10 Marks)

OR

- 4 a. Explain multithreading models with examples. (10 Marks)
- b. Explain operations on process. (10 Marks)

Module-3

- 5 a. Consider the following set of processes with arrival time:

Process	Arrival time	Burst time
P ₁	0	10
P ₂	0	1
P ₃	1	2
P ₄	2	4
P ₅	2	3

- i) Draw Gantt charts using FCFS, SJF preemptive and non-preemptive scheduling.
- ii) Calculate the average waiting time for each of the scheduling algorithm. (10 Marks)
- b. What are Semaphores? Using wait() and signal() semaphore operations for implementation, explain binary and counting semaphore with an example. (10 Marks)

OR

- 6 a. Explain how monitors can solve dining philosophers problem. (10 Marks)
- b. Explain critical section problem. List the requirements that critical section problem must satisfy. Explain Peterson's solution to critical section problem. (10 Marks)

Module-4

- 7 a. Define deadlock. What are the necessary conditions for deadlock to occur? Describe resource allocation graph: i) With deadlock ii) With a cycle but no deadlock. (10 Marks)
- b. Consider the following snapshot of a system and answer the following questions using Banker's algorithm.
- What is the content of matrix need?
 - Is the system in a safe state?
 - If a request from process P1 arrives for (1, 0, 2), can the request be granted immediately?

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3	3	3	2
P ₁	2	0	0	3	2	2			
P ₂	3	0	2	9	0	2			
P ₃	2	1	1	2	2	2			
P ₄	0	0	2	4	3	3			

(10 Marks)

OR

- 8 a. What is swapping? Does this increase the operating systems overhead? Justify your answer. (10 Marks)
- b. What are the drawbacks of contiguous memory allocation? Given five memory partitions of 100K, 500K, 200K, 300K and 600K (in order), how would each of the first fit, best fit and worst fit algorithms place processes of 212K, 417K, 112K, 426K in order? Which algorithm makes the most efficient use of memory? (10 Marks)

Module-5

- 9 a. A hypothetical main memory can store only 3 frames simultaneously. The sequence in which the pages will be required is given below:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 (20 operations)
Indicate the sequence in which the three frames will be filled in.
- FIFO
 - Optimal page replacement
 - Least recently used methods of page replacement indicate number of page faults in each case.
- Which is the most efficient among them? (10 Marks)
- b. Discuss on performance of demand paging. Explain copy-on-write process in virtual memory. (10 Marks)

OR

- 10 a. List any five typical file attributes and any five file operations indicating their purpose in one line each. (10 Marks)
- b. What are directories? Describe various directory structures. (10 Marks)

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18CS822

Eighth Semester B.E. Degree Examination, June/July 2023 **Storage Area Networks**

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define data center. List and explain the key elements of data center. Explain the characteristics of data center elements with diagram. (10 Marks)
- b. Explain with neat diagram the evolution of storage architecture. (06 Marks)
- c. With diagram explain about compute virtualization. (04 Marks)

OR

- 2 a. Explain disk drive components with diagram. (10 Marks)
- b. Discuss disk service time and disk I/O controller utilization. (10 Marks)

Module-2

- 3 a. List the different RAID levels where parity technique has been adopted. Explain any three. (10 Marks)
- b. Different between software and hardware RAID. (04 Marks)
- c. With diagram explain different RAID techniques. (06 Marks)

OR

- 4 a. Explain types of Intelligent storage systems. (10 Marks)
- b. Explain with neat diagram the components of fibre channel storage area networks. (10 Marks)

Module-3

- 5 a. Discuss different iSCSI topologies with neat diagram. (06 Marks)
- b. Briefly explain about iSCSI protocol stack. (04 Marks)
- c. Explain the components of NAS with neat sketch. Briefly explain the benefits of NAS. (10 Marks)

OR

- 6 a. Explain NAS implementations in detail. (06 Marks)
- b. Discuss about NAS file sharing protocols. (04 Marks)
- c. With diagram explain about FCIP protocol stack and FCIP topology. (10 Marks)

Module-4

- 7 a. Describe failure analysis in business continuity. Mention important BC technology solutions. (10 Marks)
- b. With a neat diagram, explain the steps involved in backup and restore operation. (10 Marks)

OR

- 8 a. What is business continuity? Explain the BC terminology in detail. (10 Marks)
- b. Discuss different backup topologies. (10 Marks)

Module-5

- 9 a. Explain FC SAN security architecture and IPSAN security implementations with diagram. (10 Marks)
- b. Explain the following :
i) Uses of local replica
ii) LVM based replication
iii) Full volume mirroring (10 Marks)

OR

- 10 a. Explain storage security domains. (10 Marks)
- b. What is remote replication? Explain storage array based remote replication. (10 Marks)

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